Sanjoy Kumar Roy

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EDUCATION

Khulna University of Engineering & Technology (KUET) April 2024
Bachelor of Science in Computer Science & Engineering Khulna, Bangladesh

- **CGPA:** 3.47 / 4.00
- **Thesis:** *Contextual Language Modeling for Bangla Using Word Embeddings* **(Grade: 4.00/4.00)**

RESEARCH INTERESTS

AI for Software Engineering, particularly AI-assisted debugging and developer tools for software maintenance.

EXPERIENCE

Samsung R&D Institute Bangladesh Ltd. Dhaka, Bangladesh
Senior Software Engineer May 2026 – Present
Software Engineer May 2024 – May 2026

Innovation & Research

- Submitted **two invention proposals** to Samsung headquarters for further evaluation and potential patent filing, following iterative local reviews of prior art and obviousness.
- Co-developed **four AI proofs of concept** in a **two-member team**, using LLMs, NLP, and computer vision.
- *Video Summarization & Contextual Search* and *Intelligent Photo Sharing App* were selected for PoC exhibition at Samsung's internal AI contest.
- Evaluated peer proposals in team innovation meetings, providing technical feedback on novelty, obviousness, and implementation feasibility.

Development

- Develop and maintain Android application-layer functionality across **five core Samsung Internet modules**: Settings, Bookmarks, Downloads, History, and Saved Pages, with responsibilities spanning UI development and database maintenance.
- Delivered updates for **three Samsung Internet market releases (v28–v30)**, supporting **One UI 8, 8.5, and 9**.
- Integrated **AI-powered features**—*Read highlights aloud* and *Translation*—into Samsung Internet's UI.
- Resolved **800+ bugs over two years**, including urgent release-blocking issues under tight deadlines, and delivered feature enhancements aligned with UX requirements.
- Collaborate with engineering teams across India, Brazil, China, Vietnam, and Samsung headquarters in South Korea to coordinate Samsung Internet feature development, integration, and maintenance.

RESEARCH & PUBLICATIONS

Contextual Language Modeling for Bangla Using Word Embeddings October 2025

- **S. K. Roy** & K. M. Azharul Hasan, *2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025. DOI: 10.1109/COMPAS67506.2025.11381858.
- Fine-tuned a pretrained **Bangla Word2Vec model** on domain-specific news text and used cosine similarity to rank candidate words for prediction.
- Compared word-prediction outputs across **three training corpora of different sizes**.

Professional Projects

Proofs of concept developed collaboratively at Samsung R&D Institute Bangladesh Ltd., with iterative review, evaluation, and refinement through internal innovation competitions.

IssueLens: AI-Assisted Issue Investigation System | **LLM, RAG** November 2025

- Co-developed an LLM- and RAG-based issue investigation system that analyzes issue descriptions, video evidence, and device logs to identify likely responsible modules and suggest probable causes.
- Implemented FFmpeg-based video frame extraction and integrated RAG over source code and feature documentation to supply relevant evidence for LLM-based issue investigation.

Intelligent Photo Sharing App | **Android, Face Embeddings, Offline P2P** June 2026

- Designed an on-device photo-grouping pipeline using MobileFaceNet embeddings and complete-linkage clustering with cosine distance, enabling person-based photo retrieval across nearby devices.
- Co-developed the application in a two-member team, implementing face detection as the primary AI contributor and supporting offline peer-to-peer integration.

Video Summarization & Contextual Search | **LLM, DeepSeek, YOLO** March 2025

- Co-developed a video analysis PoC for highlight generation and contextual search, enabling direct playback of relevant video segments.
- Implemented an AI pipeline combining DeepSeek transcript summarization with YOLO object detection, mapping summaries and detections to timestamped video segments.

TabSense: BERT-Based Website Classification & Tab Organization | **NLP, BERT** July 2025

- Co-developed a Samsung Internet proof of concept that used BERT-based topic classification to automatically organize related browser tabs into topic-based groups.
- Collected and categorized approximately **10,000 website links** through web scraping and fine-tuned pretrained BERT to classify topics from combined URLs and website descriptions.

Food Order App | **Kotlin, Android** | October 2024

- Built a complete Kotlin food ordering system with separate Android apps for customers and shop admins.
- Implemented end-to-end ordering workflows, connecting order placement with shop-side order management.

PROGRAMMING CONTESTS

- Solved **1,500+ problems** across major competitive programming platforms.
- Participated in **3 onsite** and **110+ online** contests, including an *ICPC Asia Dhaka Regional selection contest*.
- Achieved **4th place (2023)** and **8th place (2022)** as a team member at *KUET Programming Fest*.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, Python, C, C++.
Software Development: Android Development, UI and Feature Integration, Debugging.
AI & Machine Learning: LLM Applications, Deep Learning, Computer Vision; PyTorch, NLTK, OpenCV, YOLOv8.
Data Processing & Databases: Pandas, Beautiful Soup, MySQL, Firebase.
Development Tools: Git, LaTeX.

EXTRACURRICULAR & LEADERSHIP ACTIVITIES

Programming Secretary, SGIPC, KUET May 2023 – April 2024

- Organized programming contests and coordinated and conducted algorithm and data structure sessions for junior programmers through the Special Group Interested in Programming Contest (SGIPC).

Vice President, Greater Rangpur-Dinajpur Student Association, KUET May 2023 – April 2024

- Supported regional student welfare and organized admission events and the association's annual tour.